MIL-T-45398H
29 June 1983
SUPERSEDING
MIL-T-45398G
15 January 1980

MILITARY SPECIFICATION

TRUCKS, MAINTENANCE: WITH 30 TO 60 FOOT MULTI-POSITION AERIAL PLATFORM, 15,000 TO 32,000 POUNDS GVW, 4x2 AND 4X4, COMMERCIAL

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers diesel and gasoline engine driven, four-wheel, two-rear-wheel drive, and four-wheel, four-wheel drive, 15,000 to 32,000 pound gross vehicle weight (GVW), commercial trucks equipped with a maintenance body and a multi-position, articulating boom-type, aerial platform. Vehicles procured under this specification are commercial items which shall be warranted by the manufacturer as specified in procurement documents.

1.2 Classification. The vehicle shall be of the type, class, and size shown in table I, as specified (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GBS.1, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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TABLE I. Vehicle classification.

Type	Nomenclature	Class	Size				
I	Articulating boom type aerial platform with limited angle travel of not less than 125 degrees between booms.	A	30	35	--	--	--
		B	--	--	40	45	--
		C	--	--	--	--	50
		D	--	--	--	--	60
		E	30	--	--	--	--
II	Articulating boom type aerial platform with over center angle travel between booms (200 degrees minimum).	A	30	35	--	--	--
		B	--	35	40	--	--
		C	--	--	--	45	50
		E	30	35	--	--	--

1.2.1 Class. The class of the vehicle shall be determined by the GVW as specified in table II.

TABLE II. GVW ratings.

Vehicle class	GVW rating (pounds)
A	19,000
B	24,000
C	28,000
D	32,000
E	15,000

1.2.2 Size. The size of the vehicle shall be determined by the maximum height the aerial platform can attain, measured from ground level to the platform floor. Sizes shall be the minimum sizes specified in table III.

TABLE III. Sizes.

Size	Ground to platform floor height (feet)
30	30
35	35
40	40
45	45
50	50
60	60

1.2.3 Model. The model of the vehicle shall be determined by the number of the wheels and number of the driving wheels as follows:

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Model	Nomenclature
4x2	4 wheel, 2 wheel drive
4x4	4 wheel, 4 wheel drive.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.
* ZZ-T-381	- Tires, Pneumatic, Vehicular (Highway).

MILITARY

* MIL-H-46170	- Hydraulic Fluid, Rust Inhibited, Fire-Resistant Synthetic Hydrocarbon Base.
* MIL-B-46176	- Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative.

STANDARDS

FEDERAL

FED. STD. No. 595	- Colors.

MILITARY

MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Rustproofing, Identification Marking, and Data Plate Standards.
MS51118	- Pintle Assembly, Towing - 40,000 Lbs. Capacity, Manual Release.
MS51317	- Light, Warning, Vehicular, Rotating, D.C.

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

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(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)
ANSI Standards.
ANSI A 92.2 - American National Standard for Vehicle-Mounted Elevating and Rotating Aerial Devices.

(Applications for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM Standards.
ASTM Designation: 120-65 - Black and Galvanized Welded Seamless Steel Pipe for Ordinary Uses.
B117-64 - Standard Method of Salt Spray (Fog) Testing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
NFPA No. 70 - National Electric Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

SAE, INC.
SAE Standards and Recommended Practices.

	J245	- Engine Rating Code - Spark Ignition.
*	J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Trailers, and Trailers (DoD adopted).
*	J350	- Spark Arrester Test Procedure for Medium Size Engines.
	J537	- Storage Batteries.
	J551	- Performance Levels and Methods of Measurement of Elecromagnetic Radiation from Vehicles and Devices (20-1000 MHz).
	J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
	J588	- Turn Signal Lamps.
	J589	- Turn Signal Switch.
	J683	- Tire Chain Clearance - Trucks, Buses, and Combinations of Vehicles.
	J688	- Truck Ability Prediction Procedure.
*	J704	- Openings for Six- and Eight-Bolt Truck Transmission Mounted Power Take-Offs.
	J706	- Rating of Winches.
*	J816	- Engine Test Code - Spark Ignition and Diesel.
	J844	- Nonmetallic Air Brake Sytem Tubing (DoD adopted).
	J849	- Connection and Accessory Locations for Towing Doubles-Trailers and Multi-Axle Trailers.
	J1067	- Seven-Conductor Jacketed Cable for Truck-Trailer Connections.
*	J1292	- Automobile, Truck, Truck-Tractor, Trailer, and Motor Coach Wiring.
*	J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook.

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(Application for copies should be addressed to The Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.9, the vehicle, components, and accessories to be delivered under the contract as further identified and qualified in the product experience clause of this solicitation shall be the standard and optional items sold commercially to the public which meet or exceed the requirements of this specification. All chassis items shall be as represented and rated in the chassis manufacturer's technical data, and special bodies or mounted equipment shall be as represented and rated in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be on file in the engineering offices of the procuring activity, prior to delivery of the items.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Painting and marking. As specified by the procuring activity for the appropriate service (see 6.2), treatment, painting, identification marking, and data plates shall be in accordance with MIL-STD-1223, except the exterior color shall be gloss yellow, matching color chip No. 13538 of FED. STD. No. 595. Extruded, molded, or spiral wound booms and baskets shall not be painted.

3.1.1.1.1 Warning plates. Warning plates shall be furnished and mounted at a conspicuous location near the dual controls and in the driver's compartment. At least the following information shall appear on the plates:

WARNING

(a) AERIAL PLATFORM AND OUTRIGGERS MUST BE IN STOWED POSITION WHILE VEHICLE IS BEING MOVED.
(b) RATED PLATFORM LOAD, HYDRAULIC OUTRIGGERS DOWN, NOT IN EXCESS OF _____ POUNDS.

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(c) INSPECT PLATFORM LEVELING SYSTEM.
(d) MAXIMUM VOLTAGE BOOM AND PLATFORM WILL WITHSTAND _____ VOLTS AC.
(e) OVERALL TRAVEL HEIGHT WHEN BOOMS ARE IN STOWED POSITION _____.
(f) VEHICLE SHALL NOT BE OPERATED IN WINDS IN EXCESS OF 30 MPH.
(g) BASKET LINER(S) MUST BE INSTALLED WHEN OPERATING ON OR NEAR ELECTRICAL LINES.
(h) DO NOT PERMIT DIRECT CONTACT OF BASKET TO GROUND WHEN OPERATING ON OR NEAR ELECTRICAL DISTRIBUTION LINES.
(i) MAXIMUM LONGITUDINAL INCLINE _____ DEGREES.
(j) MAXIMUM TRANSVERSE SLOPE _____ DEGREES.
(k) BOOMS AND BASKET(S) MUST BE KEPT CLEAN AND FREE OF DAMPNESS.
(l) DO NOT DRILL HOLES IN BOOM OR BASKET TO DRAIN WATER, DUMP OUT WATER.

3.1.1.1.2 Instruction plates. A photo-etched, corrosion resistant plate showing a schematic of all valves, valve controls, and equipment and complete operating instructions for operating the vehicle shall be fastened on the inside of the curbside vertical cabinet door. Each control lever shall be marked with a corrosion resistant plate depicting the function of the control. Individual markings shall be adjacent to each control.

3.1.1.1.3 Warning signs. Warning signs or labels, as recommended by the Manufacturers of Aerial Devices and Digger Derrick Council, shall be furnished and attached.

* 3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with MIL-STD-1223. When specified (see 6.2), tropical rustproofing in accordance with MIL-STD-1223 shall be furnished.

* 3.1.1.3 Drain plugs. All drain plugs installed in manual transmissions, transfer cases, and rear axles shall be of the permanent magnet type.

3.1.1.4 Heavy duty cooling system. Model 4x4 vehicles shall be furnished with a heavy duty cooling system. When specified (see 6.2) for model 4x2 vehicles, a heavy duty cooling system shall be furnished. The system shall maintain engine coolant at a temperature below the boiling point with the vehicle loaded to rated GVW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F).

3.1.1.5 Towing devices. Not less than two hooks, loops, or pins for towing the vehicle shall be furnished on the front of the vehicle.

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* 3.1.1.6 Pintle assembly. A pintle assembly of the rotating type conforming to MS51118 shall be furnished. Mounting shall include reinforcement to transfer the pintle loads directly to the chassis frame rails. The rearmost portion of the pintle shall be forward, but not more than four inches forward, of the rearmost part of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength at least equal to the GVW of the truck furnished and shall be capable of accommodating a standard grab hook for a 5/8-inch chain (4-1/2 inches wide, 1-1/8 inch thick, 3/4-inch throat width). The location of the pintle assembly and the safety chain attachment devices shall be in accordance with SAE J849.

3.1.1.7 Coolant recovery system. A de-aeration system, a surge tank, or a coolant recovery reservoir with not less than a 2-quart capacity shall be furnished. A high coolant temperature or a low coolant level alarm buzzer and a red indicator warning light shall be provided on the dash instrument panel.

* 3.1.1.8 Drive hubs. When specified for model 4x4, class A and class E vehicles (see 6.2), manually operated drive hubs for disengaging front wheels from the drive train shall be provided, except for vehicles equipped with interaxle compensating devices between front and rear axles (see 3.4.5.1.2), or air brake equipped chassis.

* 3.1.1.9 Silicone brake fluid. When specified (see 6.2), and if available as manufacturer's standard or optional brake fluid, silicone brake fluid conforming to MIL-B-46176 shall be provided in the hydraulic brake system of vehicles furnished with vacuum-hydraulic or air-hydraulic brakes. A tag on the master cylinder or a permanent decal near the master cylinder shall be furnished stating "CAUTION: USE SILICONE BRAKE FLUID ONLY, MIL-B-46176."

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

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* 3.2.3 Sound level. The interior sound level shall conform to Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to EPA Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis and the cab, with all attachments, accessories, and equipment; body and aerial assembly in stowed position; and a full complement of fuel, lubricants, and coolant.

3.2.5 Gross vehicle weight. The gross vehicle weight (GVW) shall consist of curb weight, operator (weight computed at 175 pounds), and a payload distributed throughout the length of the vehicle body, to provide not less than the specified GVW. The vehicle with a crew (four-door) cab shall have GVW computed with 525 pounds of payload in the rear seat.

3.2.6 Weight distribution. The axle capacity and the capacity of the suspension and tires per axle shall be not less than as specified in table IV.

TABLE IV. Suspension, axle, and tire capacities.

Type	Class	Size	Front axle	Rear axle	GVW
I	A	30 and 35	7,000	14,000	19,000
I	B	40 and 45	9,000	18,000	24,000
I	C	50	11,000	20,000	28,000
I	D	60	12,000	20,000	32,000
I	E	30	5,000	11,000	15,000
II	A	30 and 35	7,000	14,000	19,000
II	B	35	9,000	18,000	24,000
II	B	40	10,300	18,000	24,000
II	C	45 and 50	11,000	20,000	28,000
II	E	30 and 35	5,000	11,000	15,000

3.2.7 Ratings. The vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be submitted to the engineering office of the procuring activity. Minimum GVW ratings shall be not less than specified in table II for the specified class of the vehicle. The ratings shall be established by the manufacturer of the equipment being verified.

* 3.2.8 Cab-to-axle. The cab-to-axle dimension shall not be less than as specified in table V. The cab-to-axle dimensions shall be measured from the back of the cab or crew cab to the center of the axle, as applicable.

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TABLE V. Minimum cab-to-axle dimensions.

Class	Type	Size	Minimum dimension (inches)
A	I	All	83
B	I	40	83
B	I	45	100
D	I	60	150
A	II	All	100
B	II	All	100
C	All	All	100
E	All	All	60

* 3.2.9 Overall width. The overall width of the vehicle, exclusive of mirrors, lights and reflectors, shall be not more than 96 inches.

3.3 Performance. Performance requirements for model 4x4 vehicles shall be met with the front wheel drive disengaged and with front wheel hubs (when furnished) unlocked except that vehicles equipped with interaxle compensating devices shall meet performance with the front wheel drive engaged.

3.3.1 Speeds. High and low speed requirements shall be met with the vehicle loaded to specified GVW.

3.3.1.1 High speed gradeability. The vehicle shall ascend a continous grade specified in table VI at 50 miles per hour (mph). Gradeability requirements shall be met with the main transmission in direct drive, and when a multispeed axle or auxiliary transmission is furnished, with the axle or auxiliary transmission in high speed range. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

TABLE VI. Gradeability.

Vehicle class	Percent grade
A	1.3
B	1.0
C	1.0
D	1.0
E	1.3

3.3.1.2 Low speed. Low speed shall be calculated with the engine operating at not less thn 35 percent of the recommended governed speed, and shall provide a vehicle speed of not more than 4 mph.

* 3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 60 mph (minus one percent).

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Conformance to the geared speed specified shall be determined by the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3)}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle when loaded to its specified GVW on a 30 percent grade. The service brakes shall stop the vehicle, loaded to specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

* 3.3.3 Upsetting stability. With one and one half times the specified load (see 3.6.5) placed on the aerial platform, with the vehicle otherwise unloaded, and with two outriggers extended or the spring lockouts or torsion bar actuated on sizes 30 and 35, and the four outriggers extended on sizes 40, 45, 50 and 60, the articulated booms shall be capable of placing and holding the loaded aerial platform in any attainable position of the equipment without causing the vehicle to become unstable. The vehicle shall be deemed unstable with one and one half times the specified load when any one of the vehicle wheels or outriggers lifts off the ground. The vehicle shall also be deemed unstable if with twice the specified load (see 3.6.5), the inner tire of either dual rear wheel lifts off the ground. The stability test shall not cause any permanent deformation beyond the elastic limit of any component of the vehicle.

* 3.3.3.1 Stability on slopes. With one and one third times the specified load (see 3.6.5) on the aerial platform, with the vehicle otherwise unloaded, with the vehicle on a 5-degree slope, without the vehicle being leveled, facing up, then downward, then at right angles to the 5-degree slope, and with the outriggers extended or the spring lockouts or torsion bar actuated, the articulated booms shall be capable of placing and holding the loaded aerial platform in any attainable position of the equipment without causing the vehicle to become unstable. The vehicle shall be deemed unstable with one and one third times the specified load when any one of the vehicle's tires or outriggers lifts off the ground

3.3.4 Wind resistance. With the specified payload (see 3.6.5) in the platform and in any extended position, the vehicle shall be capable of withstanding a wind load of 7 pounds per square foot without upsetting, overturning, or permanent deformation of any component. The maximum area exposure for wind load shall be calculated at the body, booms, and baskets in the most critical position that would induce tipping. The resisting moment shall be greater than the wind load when the load is applied on the surface of the body, boom, and basket. The

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contractor shall furnish calculations or a certified test report for the vehicle or identical group of the same type, class, and size of vehicles to the procuring activity as specified herein.

3.3.5 Height. The booms shall be capable of raising the aerial platform with rated load (see 3.6.5) measured from ground level to the aerial platform floor to the minimum height for the size of vehicle specified in table III.

* 3.3.6 Horizontal reach. With the lower boom in its vertical or most nearly vertical position for type I and most horizontal position for type II, the boom assembly with the upper boom horizontal shall provide a minimum horizontal reach conforming to table VII, as specified for the type and size of the vehicle, measured from the front edge of the aerial platform to the axis of the turntable. When side hung basket(s) are furnished on type I, the horizontal reach specified in table VII may be reduced not more than 1-3/4 feet. The booms of type I vehicles shall be capable of lowering the loaded aerial platform to within 2 feet of the ground for model 4x2, and to within 3 feet of the ground for model 4x4, measured from the ground to the platform floor. The booms of type II vehicles shall be capable of lowering the loaded aerial platform to ground level.

TABLE VII. Horizontal reach-upper boom.

Type	Sizes	Horizontal reach (feet)	Elevation (seconds) (min)	(max)
I	30	18	30	50
	35	21	30	50
	40	22	30	50
	45	23	40	75
	50	29	40	75
	60	34	40	75
II	30	24	30	50
	35	27	30	50
	40	31	30	50
	45	37	40	75
	50	42	40	75

* 3.3.7 Rotation (aerial platform). The boom assembly shall be capable of continuous rotation in either direction in the horizontal plane. The turntable motor shall rotate the loaded platform through an arc of 360 degrees in not less than 60 seconds and not more than 150 seconds for type I vehicles, and for type II vehicles, in not more than 230 seconds. The turntable rotating device shall remain in a fixed position when the hydraulic motor is de-energized. The turntable shall

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be capable of smoothly rotating the aerial assembly up and down a 5 percent grade when loaded to the rated working capacity, and producing the maximum attainable tipping moment.

3.3.8 Position stability (aerial platform). When the vehicle is on a 5 percent grade, without leveling the vehicle, the elevating and rotating mechanism shall hold the loaded aerial platform (see 3.6.5) in any extended position within 1 inch per hour horizontal creep and 2 inches per hour vertical settling. The creep and settling shall be measured at the front of the aerial platform with hydraulic fluid stabilized at ambient air temperature.

3.4 Chassis components.

3.4.1 Engine. The engine furnished for the specified vehicle class shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

* 3.4.1.1 Diesel engine. Unless otherwise specified (see 3.4.1.2), the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than four cylinders for class A, B and E vehicles and not less than six cylinders for class C and D vehicles. Engine net horsepower figures used in performance prediction calculations shall be determined in accordance with SAE J816 or SAE J1349. A fan clutch to reduce fan speed automatically, when not required for engine cooling, shall be provided.

* 3.4.1.2 Gasoline engine. When specified (see 6.2), the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine with not less than six cylinders. The engine furnished shall produce the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, at an engine speed not more than the manufacturer's recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel, when used in accordance with the operator's manual. Engine net horsepower figures used in performance prediction calculations shall be determined in accordance with SAE J245 or SAE J1349. A fan clutch to reduce fan speed automatically, when not required for engine cooling, shall be provided.

3.4.1.3 Oil filter. A full flow type oil filter shall be furnished.

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3.4.1.4 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the engine manufacturer's maximum recommended operating speed.

3.4.1.5 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles, the control shall include partial thermostatic control of coolant flow through the radiator and thermostatically controlled radiator shutters or complete thermostatic control of all coolant flow through the radiator.

* 3.4.1.6 Silicone rubber hoses. When specified (see 6.2), silicone rubber radiator and heater hoses shall be furnished.

* 3.4.1.7 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, an engine oil heater, and a fuel warmer (for diesel engine driven vehicles only) shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. Heaters shall operate on 120-volt alternating current (ac), and shall be wired through a junction block to a single three-pronged (male), weatherproof, slave receptacle for receiving external power and grounding vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard weatherproof, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation, and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

(a) Coolant heater, 1500-watt minimum rating, shall be installed in the engine block or lower coolant inlet hose. An engine thermostat with an operating range of 170°F to 195°F shall be installed.
(b) Immersion type engine oil heater, 300-watt minimum rating, with 170°F to 195°F thermostat, shall be installed in the oil pan through any convenient opening.
(c) Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperatures as low as minus 60°F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F.

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(d) A fuel warmer or pre-heater shall be provided on diesel engine driven vehicles to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use the engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

3.4.2 Electrical system. The electrical system shall be as specified herein, and shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system (diesel engines). For diesel engine driven vehicles, a 12- or 24-volt direct current (dc) starting system, with a 12-volt dc lighting system, and not less than a 60-ampere alternator, shall be furnished. Engine starting equipment shall include an ether starting system or glow plug. When specified (see 6.2), if an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type. Measured shot type ether systems shall be key operated or manually operated from the driver's compartment, shall have a reservoir or provisions for a reservoir of not less than 12 fluid ounces, and shall be inoperative with the engine warm.

* 3.4.2.2 Ignition system (gasoline engines). For gasoline engine driven vehicles, a 12-volt dc ignition system shall be furnished. An alternator of not less than a 60-ampere rated capacity and which provides not less than 16 amperes of dc output at normal engine idle speed shall be furnished.

* 3.4.2.3 Lighting. All vehicle lights and reflectors shall be as specified herein and shall in accordance with Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20, and 393.22 through 393.26(d).Lights and reflectors shall not be mounted on the vertical surfaces of rub rails (unless recessed and fully protected) or on the vehicle bumpers. Back-up lamps shall be furnished.

3.4.2.4 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. Vehicles shall be provided with double faced front signal lamps and with single faced rear signal lamps installed in accordance with SAE J588, except that turn signals shall not be mounted on the engine compartment hood. Turn signal operating units shall have visible and audible flash indicator.

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3.4.2.5 Lighting socket. A trailer lighting socket and cover assembly, conforming to SAE J560, shall be furnished. The connector socket shall be mounted at a readily accessible location near the pintle.

3.4.2.6 Lighting cable. A coiled electrical cable conforming to SAE J1067 shall be furnished. The cable shall be not less than 110 inches long when fully extended. Both ends of the cable shall be equipped with a seven-circuit round plug conforming to SAE J560. The plugs shall be equipped with a grip for withdrawing from the connector sockets. The cable shall be packaged and stowed in the vehicle tool compartment.

* 3.4.2.7 Electric trailer brake controller. When specified (see 6.2), the vehicle shall be provided with an electric brake controller. The controller shall be installed in the truck cab and wired through the lighting socket. The controller shall operate on the 12-volt electrical system of the vehicle and shall include load control, hand lever and accessories.

* 3.4.2.8 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity rating and the total cold cranking rating at 0°F, both measured in accordance with SAE J537, shall be not less than specified in table VIII. The batteries shall be of the maintenance free type having the maintenance free characteristics listed in W-B-131.

TABLE VIII. Batteries.

Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel	320	1200
Gasoline	100	450

* 3.4.2.9 Radio interference suppression. The vehicle shall be suppressed to limit the electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.4.3.1 Air cleaner. An air cleaner shall be furnished. When equipped with an oil-bath type, oil capacity shall be not less than 1 quart.

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* 3.4.3.2 Fuel tanks. The fuel tank(s) shall be manufacturer's standard. The vehicle shall be equipped with fuel tank(s) of not less than 20-gallon capacity for class E vehicles and not less than 43-gallon capacity for class A, B, C and D vehicles. When more than one tank is furnished on diesel engine driven vehicles, means shall be provided to assure an equalized fuel level in both tanks. When more than one tank is furnished on gasoline engine driven vehicles, a selector valve connecting either tank to the engine fuel intake shall be provided and means shall be provided to monitor the fuel level of either tank from a single fuel gage.

* 3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. The exhaust system shall be furnished with a spark arrester with an 80 percent arresting efficiency when tested in accordance with SAE J350 except on turbocharged engines. Vertical exhaust mufflers capable of being reached easily by personnel entering or leaving either side of the cab shall be provided with a heat shield. Vertical mufflers shall be furnished with a hinged rain cap.

* 3.4.4.1 Exhaust system safety shield. The exhaust system or the hydraulic system shall be shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the vehicle exhaust system.

3.4.5 Gear train.

* 3.4.5.1 Transmission. Unless otherwise specified (see 3.4.5.3), a manually shifted transmission shall be provided on the vehicle. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall be provided with a power takeoff opening conforming to SAE J704. The transmission shall provide for maximum ease of shifting in all speeds.

* 3.4.5.1.1 Speeds. The vehicle shall be provided with not less than the number of speeds specified in table IX. When more than five forward speeds are required, a multispeed transmission or auxiliary transmission shall be furnished. Gear ratios in the transmission and other gear selection equipment shall be matched to provide a progressive shifting pattern throughout the complete range, and shall provide the required vehicle performance specified in 3.3.1.

* TABLE IX. Transmission requirements, minimum number of forward speeds.

Vehicle model	Vehicle class	No. of forward speeds
4x2	A, B, C	5
4x2	D	8
4x2	E	4
4x4	A, B, C, D	10
4x4	E	8

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3.4.5.1.2 Transfer case (model 4x4). Unless the transfer case is equipped with devices which compensate for differential torque and speeds between the front and rear axles, the transfer case shall provide for driver selection of either two-wheel, or four-wheel drive. When interaxle compensating devices are furnished, they shall provide for the positive transfer of power to all the driving axles.

3.4.5.2 Clutch. The clutch shall be the largest capacity clutch offered for the type, class, and model of vehicle and engine furnished with torque capacity exceeding maximum delivered engine torque.

* 3.4.5.3 Automatic or semiautomatic transmission. When specified (see 6.2), the vehicle shall be provided with an automatic or semiautomatic transmission. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission furnished shall provide continuous drive. The transmission shall include a hydraulic torque convertor and not less than four forward gear ratios. The normal driving range selector position shall provide not less than four forward gear ratios without movement of the selector. A drive gear for the power takeoff shall be provided.

* 3.4.5.4 Power takeoff. When a power takeoff is required, the power takeoff shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver.

3.4.6 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.4.7 Frame. The chassis frame shall be the manufacturer's standard for the type of vehicle furnished. The vehicle shall have a frame with reinforcements extending at least from the rear of the front-suspension, rear hanger bracket to beyond the rear spring front hanger bracket. Reinforcements shall provide sufficient structural strength in the chassis frame through increased bending moment to at least equal the loads imposed with the vehicle body loaded to provide the specified GVW.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a total rated capacity per axle at least equal to that specified in 3.2.6. The vehicle shall be equipped with

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hydraulic, double-acting shock absorbers at the front wheels. Suspension system ratings shall be adequate for the loads imposed during all operating conditions when in a static position.

* 3.4.9 Axles. Axle ratings shall be at least equal to those specified in 3.2.6. Gear ratio(s) shall provide the performance specified in 3.3.1. Unless a two-speed axle is specified (see 3.4.9.1), a single reduction rear axle shall be furnished.

* 3.4.9.1 Two-speed axles. When specified (see 6.2), for model 4x2 vehicles, a two-speed axle shall be furnished. Axles shall be equipped with an electric, vacuum, or air shift and shall be provided with ratios which will permit proper transmission gear splitting. The gear ratio shall provide the performance specified in 3.3.1.

3.4.9.2 Traction control. When specified (see 6.2), the vehicle shall be furnished with a traction control on the rear axle. The traction control shall actuate automatically to insure that power is transmitted to the wheel(s) having traction when the opposite wheel(s) loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

3.4.10 Wheels, rims, tires, and tubes. The vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association recommendations, for the type and size of tires furnished. Tire, rim size, disc wheel size, and ply rating shall be the same for all wheels on each vehicle. When specified (see 6.2), the vehicle shall be equipped with wide base type rear tires and wheels in lieu of the dual rear wheels, and wide base type front tires and wheels. All wheels, including the spare wheel, shall be interchangeable with the front and rear axle without the use of an adapter. When specified (see 6.2), disc type wheels shall be furnished.

* 3.4.10.1 Tires. The tire capacity per axle shall be not less than as specified in 3.2.6. Steel belted radial or, when specified (see 6.2), bias ply tires shall be furnished. Tires shall be tube or tubeless type, with highway tread for model 4x2 and with non-directional mud and snow tread for model 4x4 vehicles. Tires shall conform to Tire and Rim Association Recommendations or to ZZ-T-381 with a size designation system the same as the Tire and Rim Association.

3.4.10.2 Inner tubes. For tube type tires, inner tubes shall be of the heavy duty type and shall be of proper size for the tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

3.4.10.3 Carrier for spare tire assembly. Unless otherwise specified (see 6.2), a carrier for a spare wheel or rim and tire

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assembly shall be installed in a readily accessible location on the vehicle.

* 3.4.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished. When a spare tire assembly is specified (see 3.4.10.5), a spare wheel or rim shall be furnished.

* 3.4.10.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design and load range (ply rating) as the tires furnished on the vehicle.

3.4.10.6 Tire chain clearance. Tire chain clearance in accordance with SAE J683 shall be provided. Allowance for spring deflection shall be included.

3.4.11 Brakes. Brakes shall be in accordance with Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.45 through 393.52.

* 3.4.11.1 Power hydraulic type brakes. Unless otherwise specified (see 3.4.11.2), class A and class E vehicles shall be equipped with power assisted, hydraulic actuated, four-wheel, service brakes.

* 3.4.11.2 Full air type brakes. Class B, class C and class D vehicles and, when specified (see 6.2), class A and class E vehicles, shall be equipped with four wheel, full air brakes. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head-type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 7 1/4 cubic feet per minute (cfm)
(b) Air storage reservoir(s) with not less than 2,000 cubic inches total capacity, each tank equipped with a drain cock, and with safety and check valves between the compressor and the last reservoir tank
(c) Foot control, suspended or treadle type
(d) Air control valves
(e) Air pressure gage, visible to the driver
(f) Low-air-pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Alcohol aspirator with unbreakable transparent container
(i) Automatic moisture ejector
(j) When specified (see 6.2), automatic slack adjusters.

3.4.11.3 Trailer brake control system. When specified (see 6.2), a trailer brake control system shall be furnished in addition to the

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components specified in 3.4.11.2. The trailer brake control system shall include:

(a) Identification of emergency and service lines
(b) Coincident control of trailer brakes with prime mover foot control
(c) Independent hand control for trailer brakes
(d) Towing vehicle protection valve with dash control and automatic break-away feature
(e) Trailer stoplight control operable with foot brake and with hand control for trailer brakes
* (f) Air connectors for trailers with SAE J318 gladhand couplers mounted at the rear of the vehicle, located to preclude interference with trailer
* (g) Two SAE J844 coiled air hoses, each not less than 110 inches long when fully extended, wtih SAE J318 gladhand couplers on both ends of the hoses. The hoses shall be packaged and stowed in the vehicle tool compartment for shipment
(h) Dummy gladhand couplers with chains
(i) Prime mover only parking brake valve to permit prime mover parking brakes to be applied while charging the trailer air brake system.

3.4.11.4 Brake controls for use from a towing vehicle. When specified for vehicles with air or air-hydraulic brakes (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, a relay emergency valve with no-bleed-back feature (except when a spring applied emergency brake is furnished), additional air lines, and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

* 3.4.12 Cab. Unless otherwise specified (see 3.4.12.1), manufacturer's standard full width model non-tilting cab of any type shall be furnished. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, and with at least the curbside door equipped with an external, key-operated lock. Drip rails shall be provided above the cab doors. Unless otherwise specified, the cab shall have upholstered, full-width, adjustable seat and back or individual, adjustable, driver's seat and individual passenger's seat. When specified (see 6.2), the cab shall be equipped with an individual, adjustable driver's seat and individual passenger's seat. The color of the upholstery and the interior finish shall be compatible with the

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exterior color (see 3.1.1.1). White upholstery shall not be furnished. Interior dome lighting shall be provided. Manufacturer's standard seat belt fastenings and two pairs of seat belts shall be installed.

3.4.12.1 Crew-cab. When specified (see 6.2), in addition to the requirements of 3.4.12, the vehicle shall be equipped with a four-door cab having an additional full width seat in the rear. Cab doors shall have windows with crank-operated window regulators. A rear window shall be provided. The cab roof shall be of one-piece construction or if welded, the roof shall give the appearance of one piece, with weld seams being continuous, waterproof, and free of visible seams, bumps, or protrusions. Manufacturer's standard seat belt fastenings and three pairs of seat belts shall be installed in the rear seat.

3.4.12.1.1 Seat spacing. With the front seat adjusted to the extreme forward position there shall be not less than 30 inches, measured in a horizontal plane, between the front of the rear seat-back and the rear of the front seat-back. A kick-space height of not less than 2 3/4 inches shall be maintained between the floor and the rear of the front seat in all positions of adjustment. Leg room and space forward of the front seat shall be equivalent to that provided ahead of the seat in a two-door standard cab.

3.4.13 Steering. Power steering shall be provided.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multi-speed type and operated by either air or electric motor(s).

3.4.15 Bumpers. A channel type front bumper shall be provided. Rear end protection shall be provided and shall be in accordance with Federal Motor Carrier Safety Regulation 393.86.

3.4.16 Stowage space. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, antiskid chains, and emergency reflective triangles shall be furnished for retaining the equipment during vehicle operation. Stowage space for these tools may be furnished in the cab or body cabinets. When stowage space for these tools is located outside the cab or cabinets, it shall be weatherproof and shall provide for locking with a padlock.

3.4.16.1 Tools. When specified (see 6.2), each vehicle shall be furnished with tools required for exchanging the mounted tire assembly with the spare assembly, and shall include at least a hydraulic jack, jack handle, and wheelnut wrench. The jack shall be of such closed

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height as to permit its location under the axle or other satisfactory lift point at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of the wheel and tire assembly.

3.4.17 Heater and defroster. A hot water heater shall be provided. The heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to the floor and to defroster louvers. The heater shall be complete with a blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be panel mounted.

* 3.4.18.1 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
(d) Oil pressure gage or indicator
(e) Engine temperature gage or indicator
(f) Speedometer with recording odometer
(g) Ash receptacle
(h) Dual sunvisors
(i) Front door mounted armrest on driver's side
(j) Driver's compartment ventilator other than window
(k) Tachometer.

3.4.18.2 Mirrors. Two outside rearview mirrors shall be furnished, one mounted on each side of the cab. Each mirror shall be a dual prealined type having at least 50 square inches of flat reflective area and at least 20 square inches of convex reflective area. The convex mirror shall have a radius of curvature of not less than 30 inches. The mirrors shall be inclosed in a common housing with at least three supporting arms.

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3.4.19 Horn. Manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, an air operated horn shall be furnished.

* 3.4.20 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

* 3.4.21 Front mounted winch. When specified (see 6.2), the vehicle shall be furnished with a winch assembly mounted on the front of the vehicle and powered by the vehicle's hydraulic system (see 3.6.8). The winch shall conform to SAE J706. The winch shall be of the single drum type for hoisting and negotiating purposes and shall have one forward and one reverse speed. Dual winch control shall be furnished, one inside the cab and one at the front bumper. Winch controls shall be so located as to provide no interference with the ingress or egress of the driver. The winch shall have a rated single line pull capacity of not less than 20,000 pounds on the bare drum. The winch shall be equipped with a hydraulic relief valve set to restrict the line pull capacity to 22,000 pounds. The winch line speed on the bare drum shall be between 25 to 30 feet per minute at an engine speed of 1200 rpm for manual transmissions and 2000 rpm for automatic or semiautomatic transmissions. The winch shall be wound with not less than 185 feet of 5/8-inch diameter, preformed 6x37 improved plow steel wire strand core wire rope and shall be equipped with an end chain and hook. An integral, adjustable, automatic safety brake shall be provided. The winch shall be equipped with a 4-way roller guide assembly, and shall be equipped with a shaft extension on the curb side with a capstan drive extension (AT and T type).

3.4.21.1 Winch drum guard. A winch drum guard shall be furnished. The winch drum guard shall restrict the cable to the area between the drum flanges. The guard shall consist of not less than 1/4-inch vertical side plates conforming to the outside radius of the drum flanges. Six bars, 3/8 inch by 1-1/4 inches, shall be welded to the vertical side plates. Three bars shall be located on the top and spaced equally on the top radius, and three bars shall be located on the bottom and spaced equally on the bottom radius. The vertical distance between the vertical side plates and the flanges shall be not more than one-half the specified cable diameter.

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3.4.21.2 Front bumper. When a front mounted winch is specified, the bumper shall be mounted forward of the winch. The open area on either side of the winch shall be covered with a combination step plate and gravel guard. The bumper shall be either a channel or pipe type. When a pipe type front bumper is furnished, the nominal diameter shall be not less than 3 inches, and shall have a wall thickness of not less than that specified in Schedule 40 of ASTM 120-65. The pipe type front bumper shall have half-round ball ends.

3.4.21.3 Combination step plate and gravel guard. When a front mounted winch is specified, a combination step plate and gravel guard shall be fabricated of not less than 14 gage (0.0747 inch) steel tread plate exclusive of the projections. The step plate shall be secured to the front bumper and shall be not less than 1/4 inch or not more than 3/8 inch from the chassis sheet metal. The step plate shall be capable of supporting 300 pounds per square foot. The step plate shall not deflect more than 1/8 inch under the loads imposed.

3.4.21.4 Reel. When specified (see 6.2), a reel for payout and takeup shall be provided when a front mounted winch specified in 3.4.21 is furnished. The reel shall be designed to operate from the end of the extended winch shaft. Reel yokes shall be controlled by a handle at the end of the reel shaft. The spindle shall be fitted with a bayonet type socket for attachment to the shaft. Reel spiders and yokes shall be made from heat treated aluminum. The inside dimension of the coil shall be approximately 20 inches; maximum outside diameter, approximately 28 inches; thickness or width of coil, approximately 6 inches to 9 inches.

* 3.5 Body. The vehicle shall be equipped with a line maintenance and construction type body having the minimum dimensions specified in table X for the applicable cab-to-axle. The body shall include weatherproof cabinets on each side of the center load space, hydraulically operated aerial platform assembly and turret and other equipment specified herein. Cabinet space may be utilizied to accommodate hinged type outriggers or access steps. The cabinet area that is cut to provide clearance for A-frame telescopic type outriggers shall be covered with sheet metal to present a completely inclosed and weatherproof cabinet. Safety tread steel plate shall be furnished on top of the cabinets, floor, and platform extension.

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* TABLE X. Body dimensions.

Body dimensions (minimum)	Cab-to-axle			
	60	83	100	150
Overall width of body (inch)	85	85	90	94
Loadspace width (exclusive of wheelwells (inch))	50	50	50	50
Cabinet space (each side including wheelwells):				
Length (inch) without crew-cab	106	128	144	168
Length (inch) with crew-cab	68	90	107	131
Height at front (inch) exterior	40	42	45	45
Height at rear (inch) exterior	38	40	43	43
Width (inch) exterior	14-1/2	15	15	15
Loading (without permanent deformation):				
Floor of usable load space, live load per square foot (pounds)	500	500	500	500
Cabinets and shelves, live load per square foot (pounds)	100	100	100	100

3.5.1 Bumper and platform extension. A combination bumper and body platform extension shall be provided. Safety steel tread plate shall be provided on top of the platform extension.

* 3.5.1.1 Stirrup steps. Stirrup steps shall be furnished on the combination bumper and platform extension. Grab handles shall be installed to assist personnel in entering the body. Stirrup steps and grab handles shall be furnished on the rear of both rear cabinet compartments for access to the top of the cabinets.

3.5.1.2 Access steps. When specified (see 6.2), or when a turntable is mounted rearward of the rear axle, access steps and grab handles shall be furnished on the curbside of the body, forward of the rear axle. Cabinet space may be reduced on the curbside to accommodate the access steps.

3.5.1.3 Cab protector. A cab protector to provide protection from falling objects shall be provided. The cab protector framing shall be covered with expanded sheet steel, of not less than 9 gage (0.1495 inch) prior to expansion; and not less than 90 inches in width. The cab protector shall cover the cab roof, fenders, and vehicle hood. Sufficient room above the cab roof to allow for vehicle maintenance shall be provided.

* 3.5.2 Body mounting. A subframe shall be provided. The turret, outriggers and subframe shall be one, integral, welded assembly. The subframe shall conform to the following:

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When two outriggers are required or spring lockouts are required:

(a) The cabinet-load space floor mounting shall conform to 3.5.2.2 through 3.5.5.2.3 and the subframe length shall be at least sufficient to evenly distribute the load on the chassis frame rails or:

(b) The cabinet-load space floor mounting shall conform to 3.5.2.3 and the subframe shall be the full length of the chassis frame.

When four outriggers are required:

(a) The cabinet-load space floor mounting shall conform to 3.5.2.2 through 3.5.5.2.3 or to 3.5.2.3, both with a subframe the full length of the chassis frame.

* 3.5.2.1 No welding of chassis frame. No part of the body (cabinets, floor, turret, outriggers or subframe) shall be welded to the chassis frame.

* 3.5.2.2 Flexible mounting for cabinet-floor assembly. The cabinets and the load space floor shall be an integral assembly, secured independently of the turret and the outriggers to the chassis frame or turret subframe with U-bolts, twin studs or brackets and shall inlcude a wood breaker strip. All but the rearmost cabinet-floor assembly mountings shall be spring loaded to provide for movement of the cabinet floor assembly independent of the subframe, chassis frame, derrick turret and outriggers.

* 3.5.2.2.1 U-bolts or twin studs. When U-bolt or twin studs are used for mounting the cabinet-floor assembly, there shall be not less than four U-bolts or twin studs per side having a 9/16-inch minimum body diameter with a 5/8-inch minimum thread diameter. Tie plates shall be at least 1/2 inch thick. The vehicle chassis frame and the subframe shall be braced, using wood blocks, at each mounting point unless the mounting point is located at a full depth crossmember. Blocks shall incorporate a keeper strap or groove for the mounting bolt and shall be of a width to assure retention. Two tieback straps shall be provided, one bolted to each side of the rear, to maintain cabinet-floor assembly alinement on the chassis frame rails or subframe.

* 3.5.2.2.2 Brackets. When brackets are used, they shall be bolted to the web of the subframe or the chassis frame rails. The body mounting brackets shall provide means of drawing down the body on the subframe or chassis rails and provisions shall be made to prevent lateral shifting of the breaker strips. When additional holes are required to secure the mounting brackets to the chassis frame rails, they shall be located within the area of the rail which is designated

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as being safe for drilling in accordance with the chassis manufacturer's body builders layouts. Attachments shall not interfere with nor obstruct existing chassis components.

* 3.5.2.2.3 Breaker strips. A hardwood or dense southern yellow pine breaker strip not less than 1-1/16 inches in finished thickness shall be installed between the longitudinal sills of the cabinet-floor assembly and the subframe. Breaker strips shall have a taper of 1 inch in 18 inches at the forward end.

* 3.5.2.3 Rigid mounting for cabinet-floor assembly. The cabinets and the load space floor, as an assembly or independent of one another, shall be bolted to the full length subframe. The full length subframe and the chassis frame shall be tied together in such a manner as to act as a single member resisting bending and distortion of the cabinets. The section modulus of the full length subframe and the chassis frame combined shall be not less than 70 inches cubed per frame rail. A wood breaker strip conforming to 3.5.2.2.3 shlal be provided when the cabinets and the load space floor are an integral assembly. When the cabinets are mounted independently of the load space floor, wood or metal breaker strips shall be provided when needed to provide relief at rivet heads, bolt heads, or other subframe irregularities.

3.5.3 Cabinet. Cabinet backs, partitions, and shelves shall be constructed of not less than 20 gage (0.0359 inch), high-tensile steel or 18 gage (0.0478 inch), carbon steel, reinforced. Cabinet doors shall be double panel construction of not less than 22 gage (0.0299 inch) steel. Doors shall be weather-proof and constructed to maintain fit and alinement in service. Doors shall have continuous rod or hem-type, heavy duty hinges with stainless steel hinge pins and stainless steel, nylon, or bronze insert sockets. Latches shall be furnished for all doors and shall have recessed handles. Cylinder type locks, keyed alike, shall be installed in each door. Moving parts of latches and locks shall be constructed of or plated with corrosion-resistant material. Cabinet doors over 46 inches in length or height shall have pillar bolts or two latches. Drop-down doors shall have corrosion-resistant stay chains and shall be reinforced for use as shelves capable of supporting a live load of not less than 25 pounds on any square foot of area. The cabinets and shelves shall be capable of supporting a live load of 100 pounds on any squre foot of area.

3.5.3.1 Cabinet arrangement. Vertical cabinets shall have minimum lengths of 18, 24, or 30 inches and horizontal cabinets shall have a minimum length of 49 inches, and shall be as follows:

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(a) Full height front vertical cabinets shall be a minimum of 15 inches in depth and shall be equipped with bolted on or screwed on rope hooks.
(b) Full height rear vertical cabinets shall be a minimum of 15 inches in depth.
(c) Intermediate vertical cabinets shall be a minimum of 15 inches in depth.
(d) Horizontal cabinets shall be located over the wheelwell housings and shall be a minimum of 15 inches in depth, 20 inches in height, and 49 inches in length.
* (e) Horizontal cabinets shall be equipped with small parts bins with adjustable spacers, shelves, and partitions, and a rubber goods compartment, which shall be insulated and vented. Vertical cabinets shall be equipped with not less than two adjustable tray-type shelves. The water cask when required (see 3.5.5.1) shall be mounted in the curbside vertical compartment. An insulated hot stick compartment for accommodating hot sticks at least 10 feet in length shall be furnished.

3.5.4 Lights. In addition to the lighting specified in 3.4.2.4, the following lighting and wiring shall be furnished:

(a) Lights with toggle switches and guards in each cabinet compartment.
(b) Lights on universal mounts and of sufficient candlepower to illuminate body loadspace and all equipment contained therein and all controls.

(c) Two 4-1/2 inch minimum diameter sealed beam floodlights with universal mount shall be installed, one on each side on the top of the cab protector, with controls on the outside and operable by an operator from inside the cab through an open window for illuminating the work area
* (d) Wiring for the body shall be in accordance with the applicable portion of the General Section, SAE J1292 and all wiring exposed to possible physical damage, such as wiring on a compartment floor, or wiring where tools or materials may cause the wire to be frayed or shorted, shall be protected by the use of heavy wall thermoplastic conduit or steel conduit having waterproof connections. Grommets shall be used for all wiring going through any hole or body frame work.

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(e) An amber, rotating, vehicular warning light in accordance with MS51317-4 shall be provided. The rotating light shall be mounted with reinforcement at the point of attachment on or near the left front corner of the cab guard for 360 degree visibility. The rotating light shall be operated by means of a separate switch with an indicator light located on the instrument panel and marked to indicate the function of the switch

(f) Demountable projecting load light on the end of the boom, with a service line and socket when the end of the boom extends beyond the rear of the vehicle. A receptacle for the light on the rear of the body shall be provided. The light shall be in accordance with Federal Motor Carrier Safety Regulation 393.18.

3.5.5 Attached accessories.

3.5.5.1 Water cask. When specified (see 6.2), a standard commercial 3-gallon insulated portable water cask shall be furnished. The inside liner shall be stainless steel. The outside shell shall be galvanized steel or equivalent. The throat area shall be of sufficient area to allow cleaning the interior of the water cask. A nylon or a brass chrome-plated spigot, flush mounted to preclude damage, shall be furnished. Lugs and bracket(s) shall be provided to mount the cask (see 3.5.3.1). Means shall be provided to retain the water cask and to minimize vibration during road travel. Means shall be provided to drain excess water to the ground. A cup dispenser for dispensing 6-ounce hot vending cups shall be furnished. A salt tablet dispenser shall be provided. Dispensers shall be installed in the cabinet in a location accessible from the water cask.

3.5.5.2 Vise. When specified (see 6.2), a vise shall be provided and shall have at least 3-inch hardened replaceable jaws and shall be provided with pipe jaws. The vise shall be of the swivel type and shall be bolted to the right rear corner of the platform extension.

* 3.5.5.3 Portable gasoline engine driven generator. When specified (see 6.2), a portable gasoline engine driven generator shall be furnished. The generator shall have a power rating of 120/240 volts, 60 hertz (Hz), single phase, 3-wire and provide not less than 3500 watts. The generator and its circuitry shall conform to NFPA-70. Tie down devices to secure the generator to the left front of the load space shall be furnished.

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* 3.5.5.3.1 Receptacles. The generator shall be equipped with two receptacles, one U-ground, 120-volt, 30-ampere, and one three-wire twist-lock type, 240-volt, 15-ampere.

3.5.5.3.2 Auxiliary engine for generator. The engine shall be an air-cooled, four-cycle, magneto ignition gasoline engine having not less than 7 horsepower. The engine fuel tank shall have a capacity of not less than one gallon. The fuel tank shall have a drain and an inlet valve to the engine.

3.6 Aerial tower components. Aerial tower structural components shall have a calculated design stress, such that the yield point of the materials used is not exceeded with 4 times the specified load in the aerial platform, excepting that the yield point of the plastic construction shall be not exceeded with 6 times the specified load in the aerial platform. The materials used shall meet all applicable performance requirements specified herein. All aerial devices used shall be designed and constructed in accordance with ANSI A92.2.

3.6.1 Turntable. The turntable assembly shall be rotated by a hydraulically operated motor. The turntable shall be so constructed as to eliminate all side thrusts on the motor shaft.

3.6.2 Plastic construction. Fiberglass boom(s), baskets, and components shall consist of extruded, molded, or spiral wound fiberglass, and each component shall conform to the performance requirements for electrical insulation specified herein. All components shall be from a combination and multiple layers of sterile electrical grade fiberglass roving cloth or mat with high impact strength resins with weathering and fire retardant characteristics employed. The exterior surface shall be smooth and of a uniform thickness. When gel-coat is applied, the thickness of the gel-coat shall be not less than 0.006 inch and not more than 0.030 inch. The parts shall be free from uncured areas, gaps, cracks, holes, blisters, fuzzy surfaces, resin pockets, areas lacking resin, laps, wrinkles, delamination, air or gas pockets, patches, porosity, and other similar defects. The entire surface shall be fully cured and free of tackiness. Percentage of glass by weight shall be approximately uniform. The plastic construction insulation shall meet or exceed the specified performance requirements of 3.6.3.

* 3.6.3 Dielectric properties. The upper boom shall withstand a minimum of 100,000 volts ac, 60 Hz applied across the full length of the boom for not less than three minutes without evidence of dielectric puncture and with leakage of not more than 1.0 milliampere, when tested in accordance with ANSI A92.2. Any mechanism within or along the upper

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boom shall be made of insulating materials or so insulated to demonstrate the same dielectric properties as the boom. The fiberglass basket(s) shall be completely insulated from the lower boom and turntable.

3.6.3.1 Basket(s). Single wall fiberglass basket(s) shall withstand a minimum of 30,000 volts ac, 60 Hz, without evidence of dielectric puncture (see 6.2). A decal stating "NO INSULATION VALUE WITHOUT LINER" shall be applied on each basket. The liner(s) shall withstand a minimum of 50,000 volts ac, 60 Hz without evidence of dielectric puncture. The double shell, insulated, fiberglass basket (see 3.6.5.1.1) shall withstand a minimum of 80,000 volts ac, 60 Hz without evidence of dielectric puncture.

3.6.3.2 Insulating insert, lower boom. An insulating insert shall be provided for the lower boom. The insulating insert shall withstand a minimum of 50,000 volts ac, 60 Hz, with a leakage of not more than 1.0 milliampere and there shall be no evidence of dielectric puncture.

* 3.6.3.3 Dielectric test electrode(s). Test electrode(s) shall be installed on the insulated boom section(s) in accordance with ANSI-A92.2. Test electrode(s) shall consist of two collector bands, to match the inner and outer contours of the insulated boom section(s). Collector bands shall be of a conductive material secured to the insulated boom section(s). The electrode(s) shall incorporate a terminal for test circuit connection. A dual-range meter of 0.2 milliamperes and 200 microamperes shall be furnished in a leather or heavy duty vinyl carrying case and stowed in a protected location for use by the operating personnel. An instruction plate shall be provided and shall include the maximum dielectric leakage permitted and shall describe the dielectric test procedure. The basket liner(s) and booms shall be tested in accordance with ANSI A92.2.

3.6.4 Boom assembly. The boom assembly shall be an articulated folding arm type and shall be hydraulically operated. The boom assembly shall be attached to the rotating turntable utilizing self-alining bearings and shaft. Under dynamic conditions, the boom assembly shall support the specified load placed in the aerial platform in any extended position without permanent deformation of any component in, or along the boom(s) or its supporting structure.

3.6.4.1 Boom construction. The boom(s) shall be designed, engineered, and constructed of materials that display the minimum weight conducive to the configuration and maximum performance requirements specified herein. Hydraulic lines and other associated parts which are fastened to or inside the boom(s) shall be positioned for easy

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inspection, ready maintenance or possible replacement and shall have maximum protection against damage afforded by the configuration of the booms. The boom(s) shall be attached to the aerial platform, and to the turntable by replaceable shafts. Rotating bearing surfaces of the lower boom at the turntable and at the location of the lower and upper booms which are subjected to continuous wear shall be provided with replaceable bearings or bushings. Lubrication fittings shall be provided on all rotating parts. Bearing surfaces shall be adequate to resist torsional moment when inherent with the design. When completely closed boom(s) are furnished, means shall be provided within the boom(s) to absorb moisture. Boom(s) shall be provided with inspection openings at critical areas. Boom(s) shall not be wrapped for insulation purposes.

3.6.4.2 Boom coating. The insulated boom(s) shall have a hard smooth surface inside and outside to preclude contamination and to preserve dielectric characteristics.

3.6.4.3 Boom travel of lower arm. Boom travel by the lower arm shall be a minimum of 80 degrees vertical to the body platform.

3.6.4.3.1 Boom travel upper arm. Upper arm boom angular travel for type I vehicles shall be not less than 125 degrees. For type II vehicles, the upper boom range of angular travel between booms shall be not less than 200 degrees. The cable and sheaves, when furnished, shall be compatible with each other and shall have sufficient structural strength to meet performance requirements specified herein.

* 3.6.4.4 Boom operation. The rotating turntable and boom assembly shall be hydraulically actuated by a separate mechanism and shall operate independently or in any combination of raising, lowering, and rotating at the same time. When in combination, hand metering of the controls may be necessary. The boom operating mechanism shall be capable of limiting the lifting capabilities from rest position, within safe limits of design stress to preclude overloading. For testing purposes, means shall be provided in the hydraulic system to perform the overload tests. Upon the completion of all testing, the hydraulic system pressures shall be adjusted to the manufacturer's recommended setting for the model of vehicle furnished. The vehicle shall be provided with a device to preclude the boom(s) reaching a position that would deem the vehicle unstable.

* 3.6.4.5 Boom storage. For normal storage, travel, and shipment of the vehicle, the booms shall retract to essentially horizontal positions in the vertical plane parallel to the longitudinal centerline of the

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vehicle. Saddles or a bumper mounted boom rest shall be provided to support the boom and basket assembly in the stowed position. Means shall be provided to hold down the boom assembly and platform to prevent unactuated raising of the booms when in transit. Saddles and a tie down device shall be provided to support and prevent vertical and side movement of the boom and basket assembly in transit. Boom saddles shall be sufficiently padded to preclude damage to the boom(s). The supports shall not interfere with the entrance and exit of personnel to the body loadspace.

3.6.4.6 Boom covers. When specified (see 6.2), boom cover(s) and a means to retain cover(s) on insulated sections of the booms while in stowage or stowage-in-transit position shall be provided. The cover(s) shall completely enclose the insulated arm(s). Provisions shall be made to stow the cover(s) in a compartment of the utility body.

3.6.4.7 Non-conductive extension. When specified (see 6.2), a manually operated, telescopic, fiberglass lifting extension for sizes 45 and 50 shall be provided. The extension shall be located along the side at the upper end of the upper boom and shall be provided with a lift ring or a lifting hook and spring loaded safety pin on the open side of the hook. The lifting extension shall have a capacity of not less than 1500 pounds at its most vertical position and shall conform to 3.6.3. The extension shall be marked indicating the maximum load that can be imposed on the lifting extension. A block and tackle shall be provided for the lifting extension. The block and tackle shall be provided with 1/2-inch diameter insulated rope consisting of braided polyester fibers with a braided nylon cover.

3.6.4.8 Lifting attachment for conductors. When specified (see 6.2), a lifting attachment to raise conductors off of insulators shall be furnished. The attachment shall be capable of lifting at least three conductors simultaneously by elevating the upper boom by the operator. The attachment shall be secured to the non-conductive extension. The attachment shall be provided with a mast support and a quadrant. The mast shall be of fiberglass and shall meet the requirements of 3.6.3. An insulated auxiliary cross arm, a pivoting cross arm clamp, brace, and three wire holders shall be provided.

3.6.5 Aerial baskets. The total load capacity of the aerial basket(s) shall be not less than 500 pounds for two-man capacity and 300 pounds for one-man capacity. Strength of material of the basket(s) shall be as specified in 3.6.

3.6.5.1 Fiberglass basket(s). Unless otherwise specified (see 3.6.5.2), either a single two-man fiberglass basket or two individual one-man baskets shall be furnished for sizes 40, 45, 50 and 60. A

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one-man basket shall be furnished for sizes 30 and 35. The basket(s) shall be fully enclosed on five sides. The basket(s) shall be designed with a molded-in step, protruding toward the outside, to permit ingress and egress at ground level and shall meet or exceed the performance requirements specified herein. The minimum interior dimensions shall be as specified in table XI. Minimum interior dimensions shall be measured at the bottom of the basket(s) just above the bottom radii. Non-rectangular baskets may be furnished provided the total usable floor area is equivalent to that of a rectangular basket with the minimum dimensions specified in table XI.

TABLE XI. Fiberglass basket dimensions.

Feature	Two-man basket	One-man basket
Depth to floor, inside (inches)	40	40
Width (inches)	45-1/2	21-1/2
Front to back (inches)	21-1/2	21-1/2

3.6.5.1.1 Insulated basket. When a double shell insulated basket is furnished, a basket liner need not be provided. The basket shall be constructed of a completely separate inner and outer shell of fiberglass. The space between the inner and outer shell shall be filled with polyurethane foam plastic, foamed-in-place. Dielectric strength shall be as specified in 3.6.3.1. The floor of the basket shall be provided with a removable non-conductive scuff mat.

3.6.5.1.2 Basket liner. Unless otherwise specified (see 3.6.5.1.1, 3.6.5.2, and 6.2), a non-conductive basket liner shall be provided for each basket furnished. The liner shall inclose the basket interior. The liner shall be easily removable from the basket for cleaning and testing the dielectric capacity. The basket liner shall conform to the performance requirements specified in 3.6.3.1. The floor of the liner shall be provided with a removable non-conductive scuff mat.

3.6.5.1.3 Basket covers. Waterproof basket cover(s) and means to retain cover(s) on basket(s) while in stowage or stowage-in-transit position shall be provided on each basket and also may enclose the controls; otherwise cover(s) shall be provided to completely enclose all control mechanisms. Provisions shall be made to store the cover(s) in a compartment of the utility body.

3.6.5.2 Guardrail type basket. When specified (see 6.2), a non-insulated guardrail type basket shall be furnished. The basket shall have two guardrails and a kickrail provided around the perimeter of the basket. The kickrail shall be not less than 5 inches in height and the lower and upper guardrails shall be not less than 20 inches and

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42 inches respectively from the floor of the basket. Flooring shall be of the open grating type. Interior dimensions shall be not less than 42 inches in depth, 40 inches in width and 22 inches front to back. A gate with a positive safety latch to hold the gateway closed shall be provided. Protruding edges shall be padded with a resilient replaceable bumper. The guardrail shall be constructed of 6061-T6 type aluminum.

3.6.5.3 Flooring. Basket(s) shall have a reinforced floor. No point in the flooring shall deflect more than 1/4 inch or be permanently deformed when subjected to a vertical load of 300 pounds covering an area of 25 square inches.

3.6.5.4 Safety belts. A lineman type safety belt with a positive lock and quick release feature shall be furnished for each person of basket capacity. Each belt shall have a design strength of 2,000 pounds. The safety belts shall be designed and positioned to allow freedom of movement of each workman in the basket. Means shall be provided to attach fasteners to the end of the boom or basket bracket and to preclude the fastener attachment being pulled through the plastic wall in event of a leveling system failure. Fasteners and attachments shall be corrosion resistant.

* 3.6.5.5 Tool trays. One fiberglass or polyethelene tool tray for each person of basket capacity shall be provided to fit over the edge of the basket(s). Provisions shall be made to secure and stow the trays in a compartment of the utility body.

3.6.5.6 Basket(s) engine starter and cutoff. An engine starter and cutout remotely operated from the basket(s) shall be furnished. The engine starter and cutoff shall not reduce the insulating characteristics specified in 3.6.3.

3.6.5.7 Rotating basket(s). When specified (see 6.2), rotating basket(s) having not less than 45-degree travel in either direction, transverse to the upper boom, shall be provided. Travel of the basket(s) shall be not less than 90 degrees. Gear rotation shall be self-centering in final drive and controlled from the basket(s). When two baskets are furnished, the basket with the controls may have restricted rotation in one direction to protect the controls from hitting the boom.

* 3.6.5.8 Basket leveling system. The basket(s) leveling system shall be either cable, roller chain, or hydraulic. The leveling system shall be designed with minimum stretch due to loading the basket(s). The system shall be capable of keeping the basket(s) level with the rated load in the basket(s) positioned on the floor of the basket(s).

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An adjustment shall be provided to maintain a horizontal position for the basket(s). The system shall be insulated as specified (see 3.6.3). The system shall have a minimum insulation of nine feet. Two independent leveling systems shall be furnished or the secondary system may be gravity where the bottom of the basket shall stay down in case of failure. Baskets suspended such that failure of the primary leveling system will result in gravity tipping the loaded basket shall be provided with a secondary non-gravity leveling system. The leveling system shall be automatically operated in all extended positions that the booms are capable of attaining. The basket shall be capable of being released and easily tilted over by one man, near ground level, to remove a disabled operator or to dump out water.

* 3.6.6 Controls. A complete set of hand-operated controls for maneuvering the basket(s) shall be provided both at the basket(s) and at the base of the boom. The controls shall be spaced at least four inches apart so as to permit operation with arctic gloves. Controls for maneuvering the basket(s) shall be located at the right or left side of the basket(s) in a position easily accessible to the operator facing the work area. Controls shall not be mounted on the front or rear of the basket(s). The controls shall be identified as to their function. The controls at the base of the boom shall be easily accessible from the floor of the body, otherwise a platform shall be provided at the base of the boom. Ground controls shall be placed to minimize the possibility of use by unauthorized personnel. When specified (see 6.2), a single lever control for all operations shall be furnished at the basket.

3.6.6.1 Operation. Smooth operation, with no lurching of the boom assembly shall be possible. The controls shall be self-centering and shall revert to a neutral position when released by the operator. All boom and outrigger controls shall be of the dead-man type operable with one hand but whereby two separate actions must be taken to activate any movement. The controls at the base of the boom shall be capable of overriding the controls on the aerial basket.

3.6.6.2 Mechanical or electro-mechanical control. Unless a rechargeable electrical system is specified (see 3.6.8.8), an auxiliary hydraulic pump driven by a 12-volt dc motor shall be provided for emergency use. The system shall be equipped with cut-in devices at the basket and at the lower control station. The system shall provide adequate pressure and flow capacities for operation of the hydraulic components when the primary power source fails. The system shall be capable of rotating the turntable, lowering the booms to travel position, and retracting the outriggers. The auxiliary system shall be operable from both the basket and the lower control station. A safety interlock system shall be provided to preclude operation of the booms when outriggers are not extended or spring lockouts are not actuated.

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3.6.6.3 Instruments. The switches, instruments, and pressure gages on the lower control station shall be panel mounted and identified as to their function and installed in a manner to facilitate removal and servicing. Gages and instruments shall be shock proof, waterproof and weatherproof. Identification markings shall be on a photoetched corrosion resistant plate.

3.6.7 Stabilizing equipment.

* 3.6.7.1 Outriggers. Two hydraulically operated outriggers shall be provided if the actual height (see 3.3.5) is less than 32 feet for class E and 36-1/2 feet for classes A, B, C and D. (Class E vehicle may be provided with torsion bar stabilization provided it meets all requirements of 3.3.3.) Four hydraulically operated outriggers shall be provided if the actual height is 36-1/2 feet or greater, or for class E vehicles, greater than 32 feet. Outriggers shall insure adequate stability with relation to the ground while operating the aerial platform under maximum loads. Outrigger cylinder hydraulic fittings, hoses, and pilot check valves shall be protected by metal guards and shall be accessible for service. Outriggers shall be designed so that they may be extended over curbings and sidewalks.

3.6.7.1.1 Outrigger controls. An individual control for each outrigger shall be furnished to permit leveling of the vehicle on inclined surfaces. Controls for the left outriggers shall be provided at the rear left corner of the vehicle, and controls for the right outriggers shall be provided at the right rear corner of the vehicle. All outrigger controls shall be installed in a readily accessible location which permits operation from ground level. An automatic pilot check valve shall be provided for each outrigger to block the hydraulic fluid in the cylinder and prevent movement of the outriggers in the event of a damaged hydraulic line. Interlocks shall be furnished to prevent the operation of the aerial boom assembly until the right and left side outriggers have descended and are in place on the ground surface. A self-actuating locking device shall prevent accidental raising of the outriggers while the aerial boom is elevated from a stowage position. Each outrigger shall be provided with an automatic, positive, spring actuated lock or an automatic pilot check valve which holds the outrigger in the retracted position without settling and which may be manually released.

3.6.7.1.2 Outrigger pads. Each outrigger shall be provided with ground pads. Ground pads shall be of the hinged or fixed type.

3.6.7.1.3 Outrigger warning light. A red warning light shall be located and installed in the cab in a conspicuous location to warn the driver that the outiggers are extended.

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3.6.7.2 Spring lockouts. When specified (see 6.2), hydraulically operated spring lockouts, in lieu of outriggers, shall be provided to lock out the rear springs for type I, sizes 30 and 35 only. Means shall be furnished in the interlock system to prevent the boom assembly from elevating until the spring lockouts have been actuated. When spring lockouts have been retracted, they shall not interfere with the spring characteristics during transit. With spring lockouts actuated, the vehicle shall meet the performance requirements specified in 3.3.3 through 3.3.8.

3.6.7.3 Circular level. A circular level, with 0 to 5 degrees reading, shall be furnished. The circular level shall be mounted in a protected and visible position at the outrigger or spring lockout controls. The circular level shall indicate the degree of body level when the outriggers are extended. The circular level shall be corrosion resistant and resistant to inclement weather. A warning plate (see 3.1.1.1.1 and 3.1.1.1.2) shall be mounted adjacent to the level to indicate the maximum longitudinal incline and the maximum transverse slope in degrees on which the vehicle is capable of operating.

3.6.8 Hydraulic system. The hydraulic fluid non-conductive system shall consist of the components required to perform all hydraulic operations of the boom assembly, spring lockouts, and outriggers with the degree of control specified herein. The hydraulic system shall also include the components required to power the front mounted winch, when furnished (see 3.4.21). Mechanisms and reservoirs shall be positioned or shielded to prevent hazardous fluid contact with heated surfaces or electrical apparatus. A two-speed manual or an automatic hydraulic throttle actuator shall be furnished which increases the speed of the auxiliary or truck engine when the pump is being loaded. An accumulator may be used as a component of the hydraulic system to provide the maximum degree of temperature control under continuous service.

* 3.6.8.1 Hydraulic cylinders. All hydraulic cylinders shall incorporate seals to prevent entrance of moisture and dirt into the cylinders and leaks of hydraulic fluid from the cylinders. Preformed packings shall be used in the cylinders where static seals are required. Manufacturer's standard seals shall be furnished where applied to reciprocating parts. All hydraulic cylinders for actuating the booms, spring lockouts, and outriggers shall be provided with pilot check or holding valves to prevent descent of the boom assembly, and retraction of the outriggers or spring lockouts in case of loss of hydraulic pressure. Means shall be provided to install pressure gages for testing the hydraulic system pressure and relief settings. All surfaces of hydraulic ram piston rods shall be chrome plated with sufficient thickness to withstand a 30-hour salt spray test in accordance with ASTM B117-64.

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3.6.8.2 Hydraulic lines and fittings. Hydraulic lines shall be located to facilitate maximum ease of removal and installation, or visual inspection for failures and maximum protection from damage during operation of the boom assembly. Flexible, hydraulic hose or collector rings shall be used where flexing of hydraulic lines is encountered and where installation of components requires a flexible coupling. Standard straight-thread, flared type fittings or adaptions to flared type fittings, or tapered pipe thread with dry seal fits shall be furnished. Hydraulic lines and fittings shall be air and liquid tight and shall be installed to preclude vibration and chafing. The system shall be designed to preclude hydraulic transients. Provisions shall be made for bleeding air from the hydraulic system and completely draining all hydraulic lines, when necessary. A manual type bleeder shall also be provided as on-vehicle equipment on closed-center systems. Hose assemblies and all seamless steel hydraulic lines shall be adequate to provide the rate of flow, temperature, and pressures required to operate the system smoothly and efficiently. Galvanized pipe and copper tubing shall not be acceptable. A pressure gage shall be provided on the discharge side of the hydraulic pump.

* 3.6.8.3 Hydraulic pump system. A hydraulic pump with necessary pressure and flow capacities to operate all hydraulic powered components shall be provided. Unless otherwise specified, the hydraulic pump shall be driven by a heavy duty power takeoff unit attached to the truck chassis transmission. When specified, the hydraulic pump shall be driven by the rechargeable electrical system (see 3.6.8.8). When specified (see 3.6.9), the hydraulic pump shall be driven by an auxiliary engine. When an air compressor is specified (see 3.6.10), the hydraulic pump and the air compressor shall be driven by an auxiliary engine. Chain drives shall not be used. The pumping system shall include an adjustable pressure relief valve or unloading valve with the range of adjustment which allows raising or lowering of the boom(s) and basket(s) with any working or test loads specified herein.

* 3.6.8.4 Hydraulic fluid reservoir. A hydraulic fluid reservoir shall be provided and shall have sufficient capacity to hold the required volume of hydraulic fluid, with not less than 10 percent for expansion space. The reservoir shall be completely baffled. The reservoir shall be provided with a filler opening and removable oil strainer and vented filler cap, accessibly located to facilitate filling the tank from a one-gallon container using a funnel. The vented filler cap shall have means to preclude fluid from splashing out of the tank during movement of the vehicle. The filler cap shall be attached to the tank by a corrosion resistant chain to prevent loss. Provision shall be made for inspection and cleaning of the interior and draining of the reservoir. The reservoir drain shall be so located that the fluid will

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not drain on other components of the vehicle, and shall allow recovery of the fluid in a container. The reservoir shall have a photoetched, corrosion resistant plate to indicate the type of hydraulic fluid furnished, and the maximum fluid level in the reservoir with the booms folded for travel position and the hydraulic lines and components full of fluid. A sight level or dip stick gage shall be furnished to observe the level of fluid. The tank shall be clean.

* 3.6.8.5 Hydraulic fluid. Unless otherwise specified, hydraulic fluid conforming to the manufacturer's standard or recommended requirements shall be provided in the hydraulic system. When specified (see 6.2), hydraulic fluid conforming to MIL-H-46170 shall be provided in the hydraulic system.

3.6.8.6 Oil filter and strainers. A full flow type 10-micron discharge oil filter in the return line to the reservoir shall be provided. A 100-mesh suction line strainer in the hydraulic reservoir shall be provided. Filter and suction strainer shall be easily accessible for periodic cleaning, or for replacement of the filter cartridge. Filter and strainer shall have sufficient capacity to allow the maximum rated pump discharge.

* 3.6.8.7 Hydraulic fluid temperature. The hydraulic fluid temperature shall be maintained within safe limits to preclude seal damage or approach to flash point of the fluid during all operating conditions of the vehicle and use of the aerial tower for long periods of time. Excessive heating (more than 55° F above ambient) of the fluid shall not be acceptable under any operating condition. Provisions shall be made to prevent malfunction or unactuated operation of the hydraulic control system due to changes in fluid temperature.

3.6.8.8 Rechargeable electrical system. When specified (see 6.2), a rechargeable electrical system shall be furnished to drive the hydraulic pump assembly. The electric motor shall be mounted on the body platform in conjunction with the hydraulic pump. The mounting shall minimize the transfer of vibration. The electrical system shall include:

(a) 36-volt dc motor
(b) 375 ampere-hour industrial battery charger
(c) 220-volt, single phase battery charger
(d) Hourmeter
(e) Battery charge indicator
(f) Pressure switches and control circuits
* (g) Connection of all vehicle lights to the rechargeable electrical system to enable operation of the lights when the system is being used.

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* 3.6.9 Auxiliary engine. When specified (see 6.2), an auxiliary engine shall be provided to drive the hydraulic pump assembly. When an air compressor is specified (see 3.6.10), an auxiliary engine to drive the hydraulic pump and the air compressor shall be furnished. The auxiliary engine shall be air cooled, and shall operate on the same fuel as the vehicle chassis engine. The engine shall be mounted on the body platform. The engine mounting shall minimize transfer of vibration. During normal operating conditions, the continuous-duty horsepower output required by all components and accessories driven by the engine shall not exceed 75 percent of the full throttle rating of the engine. The size and weight of the engine assembly shall be held to a minimum consistent with the performance requirements specified herein. The engine shall be capable of continuous warranted operation on unleaded fuel.

3.6.9.1 Engine components. The engine shall be a two or four-cycle diesel engine or a four-cycle gasoline engine equipped with at least the following components:

(a) Distributor or magneto type ignition (gasoline engine only)
(b) Self-starter, 12-volt
(c) Resistor-type spark plugs (gasoline engine only)
(d) Oil bath air cleaner
(e) Automatic choke (gasoline engine only)
(f) Engine hour meter conforming to 3.4.20 but for the auxiliary engine
(g) When specified (see 3.4.1.7), power plant heater having a 300-watt immersion type heater with 170°F to 195°F thermostat shall be installed in oil pan through any convenient opening

3.6.9.2 Engine control panel. The engine accelerator, key-operated starting (ignition) switch, and palm button starter shall be installed in a weather and waterproof cabinet on a control panel, and each shall be identified as to their function. The cabinet shall be provided with a side-hinged cover with a spring-type latch and provisions for locking with a 1-inch padlock.

3.6.9.3 Electrical system. The electrical system shall utilize the truck-chassis battery and shall be in accordance with Federal Motor Carrier Safety Regulations 393.30 through 393.33.

3.6.9.4 Fuel pump. A fuel pump shall be furnished on the engine and when lifting fuel against a head of 4 feet, the pump shall have sufficient capacity to supply the engine at its maximum rated load. An in-line gasoline fuel filter shall be provided.

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3.6.9.5 Governor. A variable speed governor shall be furnished. The governor shall automatically control the engine speed at the selected rated speed at all fluid pressures and flows required. A manually operated hand throttle shall control the engine at all speeds from the manufacturer's recommended minimum governed speed up to the maximum operating speed.

* 3.6.9.6 Auxiliary engine fuel supply. Unless an auxiliary engine fuel tank is specified (see 3.6.9.6.1), the auxiliary engine shall utilize the vehicle's main engine fuel tank(s) (see 3.4.3.2). A shut-off valve shall be provided in the line to the power source and shall be located in an accessible location. Fuel line(s) shall be in accordance with Federal Motor Carrier Safety Regulations 393.65 and 393.67.

* 3.6.9.6.1 Auxiliary engine fuel tank. When specified (see 6.2), an auxiliary engine fuel tank shall be furnished. The fuel tank shall have a capacity to provide fuel for at least 8 hours of full-load operation. The fuel tank cap shall be attached to the tank by means of a corrosion resistant chain. A sump shall be provided in the fuel tank suitable for collection and accumulation of water. A drain shall be located at the bottom of the sump for complete drainage of the fuel tank. Drainage shall be provided to preclude spillage of fuel on any area of the vehicle. The tank filler shall be provided with a protective screen and vent. A shut-off valve shall be provided in the line to the power source and shall be adjacent to the fuel tank. A photoetched corrosion resistant plate shall be furnished and marked denoting the type of fuel to be used and shall be located in a conspicuous location near the filler cap.

3.6.9.7 Exhaust system. The exhaust system shall be furnished with a spark arresting muffler and means shall be provided to vent exhaust fumes away from the cab and body.

3.6.9.8 Exhaust system safety shield. The exhaust system or hydraulic system shall be shielded so that, in the event of a hyraulic line rupture or hydraulic fliud leak, the hydraulic fluid will not come into contact with any part of the vehicle exhaust system.

3.6.10 Air compressor. When specified (see 6.2), an air compressor shall be furnished. An auxiliary engine conforming to 3.6.9 through 3.6.9.8 shall be provided to drive both the compressor and the hydraulic pump. The compressor shall be the air-cooled two-stage type with constant speed control, shall have an actual displacement of 30 cubic feet per minute (cfm), and shall provide a pressure of not less than 150 pounds per square inch (psi). The air

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compressor shall be complete with receiver, moisture separator, automatic oilers, shock and waterproof pressure gage, safety valve, and all necessary equipment for satisfactory operation of the compressor. An air line extending from the compressor to an outlet in the cabinet at the curbside of the vehicle shall be provided. An air line extending from the compressor to outlets at the basket(s) shall be provided and shall be of the non-conductive type. An air line shut-off valve shall be provided at the basket outlet(s). The compressor shall be so installed as to minimize the transfer of vibration from the auxiliary engine. The receiver tank shall have a capacity of not less than 30 gallons. An air pressure regulator having a setting range of at least 50 to 150 psi shall be furnished.

3.6.10.1 Pneumatic outlet. An air outlet shall be furnished in the basket. One 8-foot air hose, for use in basket, shall be furnished. The hose shall be rubber non-conductive with quick disconnect fittings, compatible with the outlet and with standard pneumatic tools. Both ends of the air hose shall be provided with dust caps. All critical pneumatic components shall comply with ANSI Standard A92.2.

3.6.11 Hydraulic tools.

3.6.11.1 Hydraulic tool outlet. A hydraulic tool outlet shall be provided in the curb side cabinet or in a protected position on the rear or under the platform. Discharge port(s) and return port(s) shall be provided with a corrosion-resistant cap, secured with a corrosion resistant chain to protect the threads and the valve from contamination. Flow divider control valves shall use the hydraulic fluid from the hydraulic system of the vehicle. One tool circuit with quick disconnect, dripless couplers shall be furnished for attaching the pressure and return hose lines for the motor driven hydraulic tools. The oil flow for the tools shall be regulated by control valve(s) to meet the flow requirements of the hydraulic tools. The maximum pressure shall be 1,700 psi to 1,800 psi and the back pressure in the return line shall not exceed 150 psi.

3.6.11.2 Basket hydraulic tool outlet and hose. Unless otherwise specified (see 6.2), a hydraulic tool outlet shall be furnished in the basket. An 8-foot dual hydraulic hose, for use in basket, shall be furnished. The hose shall be a rubber, non-conductive type with quick disconnect, dripless fittings on each end. Hose fittings shall be compatible with the basket outlet and the tools specified in 3.6.11.4 through 3.6.11.9. The hose ends shall be provided with dust caps.

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3.6.11.3 Ground tool hose. Except when a hydraulic hose reel is specified (see 3.6.11.12), two 50-foot lengths of hydraulic wire braid hoses with quick disconnect, dripless couplers on both ends for use on ground level shall be furnished. Instruction plate markings (see 3.1.1.1.2) shall indicate that the wire braid hydraulic hoses shall not be used on or near electrical lines. The hose shall be furnished with dust caps. Hose fittings shall be compatible to the tool outlet(s) specified in 3.6.11.1 and the tools specified in 3.6.11.4 through 3.6.11.9.

3.6.11.4 Hydraulic capstan hoist. When specified (see 6.2), a hydraulic capstan hoist shall be furnished. The drum shall be one direction drive and shall have a backstop on the output shaft to prevent reverse travel. Lowering will be accomplished by letting the rope slip on the drum. The capstan hoist shall be complete with a link chain and screw clamp attachment device for mounting to surfaces other than on a round pole. The hoist shall have a rated single line pull capacity of not less than 1,250 pounds on the bare drum. The hoist shall have a remote control foot operated open-center valve.

3.6.11.5 Hydraulic impact wrench and drill. When specified (see 6.2), a hydraulic impact wrench and drill shall be furnished. This combination tool shall be reversible and capable of installing or removing lag screws and other hardware. The wrench shall have a 5/8-inch standard bolt size capacity with a 1/2-inch square drive. The tool shall be capable of developing not less than 200 foot-pounds of torque with at least 1,350 impacts per minute. A quick change drill chuck shall be furnished.

3.6.11.6 Hydraulic chain saw. When specified (see 6.2), a hydraulic chain saw shall be furnished. The saw shall have a chipper-type chain and shall be suitable for cutting trees up to 15 inches in diameter.

3.6.11.7 Hydraulic chain saw, pole mounted. When specified (see 6.2), a hydraulic chain saw, pole mounted, shall be furnished. The chain saw shall be light weight (approximately 10 pounds) and shall have a cutting capacity of 9 inches. The pole shall be approximately 5 feet long and shall have a fiberglass type of body to provide good insulation.

3.6.11.8 Hydraulic circular saw, pole mounted. When specified (see 6.2), a hydraulic circular saw, pole mounted, shall be furnished. The saw shall have a capacity of 3-3/4 inches and shall operate at

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approximately 8900 feet per minute. The pole shall be approximately 5 feet long and shall have a fiberglass type of body to provide good insulation. The saw shall have at least a 40- and an 80-tooth blade.

3.6.11.9 Hydraulic pruner, pole mounted. When specified (see 6.2), a hydraulic pruner, pole mounted, shall be furnished. The pruner shall have a capacity of 2 inches in diameter. It shall be mounted on a fiberglass type pole to provide good insulation and shall be approximately 7 feet long.

3.6.11.10 Hydraulic fittings. Quick disconnect, dripless fittings shall be provided on each hydraulic tool. The fittings shall be compatible with the hoses specified in 3.6.11.2, 3.6.11.3 and 3.6.11.12.

3.6.11.11 Hydraulic tool stowage. Hydraulic tools, when furnished, shall be stowed on brackets inside a curbside cabinet. The brackets shall be provided with retaining devices to secure the tools.

3.6.11.12 Hydraulic hose reel. When specified (see 6.2), a heavy duty, retractable hydraulic hose reel and hose shall be furnished. The hose reel shall be furnished with a mounting plate secured in the vehicle cabinet or in an accessible location on the vehicle platform. The hose reel shall be furnished with a weatherproof cover for protection when not in use. The hose reel shall be capable of withstanding not less than 1,800 psi line pressure. The hose and hose reel shall be capable of meeting the dielectric properties specified in 3.6.3. The hose reel shall be capable of retraction. The hose reel shall be marked stating that the hose shall be walked back to the reel to preclude damage. Hose fittings shall be of the quick-disconnect, dripless type and shall be compatible with the tool outlets specified in 3.6.11.1 and 3.6.11.2 and with the tools as specified in 3.6.11.4 through 3.6.11.10. The following accessories shall be furnished:

(a) Flow control valve(s)
(b) Line check valve(s)
(c) Hose ball stop
(d) Four-way roller hose guide
(e) Dual hose, 50 feet in length, with dust caps.

3.6.12 Safety. All equipment or exposed portions of the equipment which are subject to extreme temperatures and inclement weather, and all rotating or reciprocating parts which are of such a nature or so located as to become a hazard to operating personnel, shall be insulated, fully enclosed, or properly guarded.

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3.6.13 Highway warning kit. Three emergency reflective triangles conforming to Federal Motor Carrier Safety Regulation 393.95 (f)(2)(i), shall be furnished and stowed in the cabinets.

3.6.14 Stress analysis. A stress analysis for each type, class, and size vehicle produced shall be submitted to the procuring activity. The stress analysis shall include the weight of all major items permanently installed on or within the vehicle, and shall show the location of the center of gravity of each component relative to the vehicle axles (travel and critical working positions). In addition, the data shall include the value of stresses at the base of the boom, at the hinge area between the upper and lower boom sections, at the hinge area between the upper boom section and the basket(s), and the center of gravity of each section of the boom. The analysis shall verify the factors of safety specified herein, and shall verify that the yield point of the material used is not exceeded with the specified load on the basket(s). Stress analysis shall be provided for each vehicle chassis model and body model and size.

3.7 Dissimilar metals. All dissimilar metals throughout the vehicle shall be insulated from one another to preclude galvanic action or electrolytic action.

3.8 Lubrication system. A lubrication sytem shall be furnished with fittings for all bearings and other working parts which require relubrication for continued service. Vehicles shall be lubricated, prior to delivery, with an applicable grade of standard lubricant for the ambient air temperature encountered at the delivery point.

* 3.9 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: alinement of lights; adjustment of the engine and brake systems; filling and charging of batteries; alinement of front wheels; inflation of all tires; complete lubrication of chassis, engine, running gear, and mounted equipment with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of windshield washer reservoir with water and appropriate additives. Cables, where used, shall be lubricated with proper lubricants to resist corrosion.

3.10 Workmanship. The quality and workmanship of all component parts, including fabricated parts, machined parts, and castings shall be commensurate with sound manufacturing standards for similar equipment. The vehicles shall be free from all burrs, sharp edges, die, tool and

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metal handling marks, rust, scale, and cracks. All metal and plastic materials used shall be free from delamination defects, and shall be prime material. Improper manufacturing or assembling practices, defective components or parts and assemblies which have been modified to overcome deficiencies shall not be furnished.

3.10.1 Welding. Adequate preparation of metal prior to the welding of the body and aerial tower components shall be provided. The surface of parts to be welded shall be free of rust, scale, paint, grease, and other foreign matter. Welding rods shall be compatible to the metal being used. All critical arc welds shall be magnaflux or dye penetrant inspected. The following defects shall not be acceptable:

(a) Welds which are ground flush with metal
(b) Slug or plug welding
(c) Undersize of fillet welds, overlaps, inadequate fusion and penetration, undercutting of weldments, and vertical down welding
(d) Unremoved slag and spatter
(e) Cracks
(f) Spot, tack, or intermittent welds for strength.

Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Intermittent welding shall be permissible when in accordance with 3.10.2 through 3.10.3.

3.10.2 Workmanship specimens. Workmanship specimens shall be prepared by the manufacturer for the body and aerial tower components to simulate all welding conditions. The specimens shall represent the minimum acceptable weld quality and cleaning procedure to be used in production. Upon submittal of these specimens by the contractor, the quality level and identification shall be subject to approval by the Government. The specimens shall be prepared using the welding procedures to be employed in production. Specimens prepared to represent multiple pass welds shall be made in such a manner as to leave at least two inches of each layer of weld metal exposed. Approval specimens shall be retained at the contractor's plant.

3.10.3 Welding procedures. The welding procedure shall be recorded prior to production of any weldment, but no procedure qualification other than workmanship specimens shall be required.

3.10.4 Factors. The factors listed below which are applicable shall be included in the welding procedures:

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(a) Joint sketch
(b) Joint type and preparation
(c) Base metal type
(d) Filler metal, size and type
(e) Position of welding
(f) Arc-voltage, current range and polarity
(g) Manual or machine
(h) Travel speed, machine only
(i) Pre-heat and post-heat.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First vehicle preliminary inspection.

4.3.1 Welding procedure and workmanship specimen evaluation. Prior to the start of fabrication of the subfloor, body, turntable, outriggers, and boom assemblies, an inspection shall be conducted by the Government to evaluate workmanship specimens and welding procedures.

4.3.1.1 Macro-etching. Workmanship specimens shall be cross-sectional and macro-etched. Both parts of the specimens shall be made available to the Government representative for examination.

4.3.2 Radio frequency suppression verification. The vehicle manufacturer shall furnish, with the questionnaire (commercial vehicle engineering data), a description of all interference suppression devices

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used to suppress vehicle electromagnetic radiation in accordance with SAE J551. Data shall be sufficient for visual determination that all required suppression devices are installed on the vehicle.

* 4.3.3 In-process inspection of body and components. The first completely assembled body and components (subframe, body, turntable, outrigger, boom assemblies, and basket) shall receive an in-process inspection by the Government prior to prime painting. The inspection shall be conducted to evaluate workmanship and to determine conformance of structural members to contractual requirements.

* 4.3.3.1 Structural inspection. The vehicle shall be tested in accordance with 4.4.7 and 4.4.8 and dye penetrant or magnaflux shall be applied on the critical areas of the weld prior to painting. A water soluble dye penetrant shall be applied on all critical plastic areas. The water soluble dye penetrant or magnaflux shall provide a means to inspect for crack indications under dynamic and static conditions.

4.4 First production vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.4.1 Weight and payload capacity. The first production vehicle shall be weighed to determine curb weight and distribution of curb weight on front and rear axles. The imposed loading on the front and rear axles shall be computed to ascertain that the vehicle is adequate for the intended loading. The distribution of weight shall be computed by using curb weight, the operator weight at 175 pounds, and a payload considered to be evenly distributed throughout the length of the vehicle body, to provide the specified GVW.

4.4.2 Road test. The first production vehicle shall be examined and road tested by the supplier to assure that the vehicle shall operate in accordance with contractual requirements. The vehicle loaded with payload and all the equipment specified herein, shall be driven at speeds up to at least 50 mph, for a minimum distance of 50 miles. The distance traveled shall consist of highway, rough dirt roads, partly surfaced gravel roads, puddles, and sharp turn. For each 10 miles of travel at least one sudden stop shall be accomplished. Upon completion of the road test all equipment shall be thoroughly examined. The equipment shall show no evidence of damage, all compartment and cab doors shall be opened and shut, there shall be no evidence of mud or dust inside, and there shall be no evidence of racking of the body.

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4.4.3 Upsetting and range of movement test. With the truck without payload resting on level ground and not fastened to any artificial base, the basket(s) shall be loaded with 1,000 pounds for two-man capacity and 600 pounds for one-man capacity. The basket(s) shall be extended to its full reach and exercised through the full range of traverse and elevations indicated in 3.3.3, for type I and type II vehicles and with the limiting device on type I and II vehicles, and shall include at least the following positions:

(a) When the upper and lower arms are in a horizontal or in its most horizontal plane, and extended to maximum reach
(b) When the lower arm is horizontal or near horizontal and to the side of vehicle and the upper arm in most vertical position possible
(c) When the lower arm is at maximum travel from the stowed position, and the upper arm horizontal and 45 degree to the side of the vehicle or over the four corners of the vehicle. Rotate turntable with twice the specified load (see 3.6.5) 360 degrees and stop each 15 degrees. Reverse the turntable and stop each 20 degrees. The inner dual tire shall be observed for conformance to the stability requirements of 3.3.3 with twice the load (see 3.6.5). The test shall be repeated with one and one-half times specified payload (see 3.6.5) and the outriggers and all tires observed for conformance to the stability requirements of 3.3.3 with one and one half-times the load.

* 4.4.4 Wind test. With the vehicle not loaded, with the outriggers extended or spring lockouts actuated, and the rated capacity placed in the basket(s), the aerial tower shall be extended to maximum height and shall be rotated to that position whereby the greatest area of the basket(s) is parallel to the longitudinal centerline of the vehicle. The specified wind load, generated by artificial means or by an equivalent simulated load, shall be applied to the vehicle from either side (see 3.3.4). The contractor shall furnish calculations or a certified test report to the procuring activity.

4.4.5 Speed test. With the vehicle resting on approximately level ground, the basket(s) shall be loaded with specified load (see 3.6.5). The basket(s) shall be traversed and elevated to determine conformance to the requirements in 3.3.5, 3.3.6, and 3.3.7.

* 4.4.6 Creep test. With the vehicle longitudinally on a 5 percent grade, without leveling the vehicle, and with specified load (see 3.6.5)

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on the basket(s), the basket(s) shall be extended to the side with the lower arm in vertical or nearest capable vertical position and the upper arm in horizontal position. The position shall be held for at least 1 hour for maximum horizontal creep of 1 inch per hour and vertical settling of 2 inches per hour, to determine conformance to requirements in 3.3.8.

4.4.7 Dynamic conditions. With two times the specified payload (see 3.6.5), on basket(s) floor, the upper and lower booms shall be extended through various positions, as deemed necessary by the Government inspector, and with the turntable being rotated 360 degrees, for at least 1/2 hour, the water soluble penetrant and magnafluxed areas shall be inspected for any evidence of cracks. Any evidence of cracks or deformation shall be cause for rejection of the vehicle.

* 4.4.8 Static conditions. With the lower boom in its most vertical position, and the upper boom horizontal, and with 2-2/3 times the specified load (see 3.6.5) placed in or suspended from the basket(s), water soluble dye penetrant and magnafluxed areas shall be inspected for any evidence of cracks. Any evidence of cracks or deformation shall be cause for rejection of the vehicle. Dye penetrant and magnaflux inspection shall be conducted prior to the first production vehicle inspection. The tests in 4.4.7 and 4.4.8 shall be repeated at the first production vehicle inspection; and records of the tests and dye penetrant and magnaflux inspection witnessed by the Government shall be presented at the time of the first production vehicle inspection.

4.4.8.1 Critical areas. At least the following critical areas shall be inspected by dye penetrant or magnaflux:

(a) Outriggers, spring lockouts, and attaching areas
(b) Turntable, base, trunnion, and yoke at attachment of lower boom and struts
(c) Lower boom at cylinder clevis brackets
(d) Lower boom sheaves, sheave supporting brackets, cable anchor, and support brackets
(e) Lower boom at attachment to upper boom
(f) Upper boom sheaves, sheave supporting brackets, cable anchor, and support brackets
(g) Structural steel of upper and lower boom
(h) Upper boom and yoke at attachment to basket(s) and side bracket.

4.4.9 Dielectric tests. The following dielectric tests shall be conducted at line-to-ground voltages specified in 3.6.3. The dielectric test circuit shall be tested (see 3.6.3.3). For all tests except the basket immersion test, all components shall be assembled for operational use prior to the tests.

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(a) Vehicle under and parallel to and not more than 12 inches away from an energized conductor bar, with ground cable attached to the vehicle, and to test unit. Apply required voltage when conductor bar is not attached to vehicle. The total internal leakage in milliamperes will be indicated on the test unit and shall be recorded

(b) Secure a cable from the energized conductor bar to the following areas and apply the required voltage. The total leakage in milliamperes will be indicated on the test unit:
  (1) Basket bottom through the insulated booms to ground, with and without basket liner
  (2) Basket, front, through the insulated booms to ground, with and without basket liner
  (3) Basket attachment, through the booms to ground, with and without basket liner
  (4) Boom, upper, at hinge point to lower and upper boom to ground (only when lower boom is insulated)
  (5) Top of upper boom to upper and lower boom hinge area to ground

(c) Disconnect ground cable and attach to basket and test as follows:
  (1) Secure a cable from the energized conductor bar to the basket front and test, then the bottom, with and without basket liner, and apply the required voltage. The total leakage in milliamperes will be indicated on the test unit, and shall be recorded for each test

(d) Basket immersion test, with basket immersed in a brine solution and interior of basket filled to a depth that is practical to preclude arcing

(e) Foamed-in-place fiberglass basket (see d).

4.4.10 Performance test. The platform shall be subjected but not limited to the following tests.

(a) Check to determine whether boom assembly can be elevated without outriggers for type I and II vehicles and when lockouts are specified

(b) Determine if outriggers have locked automatically when in an extended position and in a retracted position and spring lockouts have been actuated

(c) Operate equipment using each set of controls separately

(d) Check to insure that the lower controls override the basket controls

(e) Check to insure that the entire system can be operated smoothly and without jerk motions in any position

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(f) Check to determine whether the basket(s) remain level in all positions with twice the specified loads in the basket(s)
(g) Check to determine whether the basket(s) are safely lowered from maximum reach position with rated payload on the basket(s) floor to within 36 inches above the ground, with the hydraulic pump and engine not operating, except as specified in 3.3.6
(h) Check the remote control starter and cutoff on basket(s), to insure starting and stopping engine with a specified payload on the basket(s)
(i) Check overload capabilities by attempting to lift in excess of two times the rated capacity of the basket(s) without increasing hydraulic pressure (see 3.6.5) with lower boom vertical and upper boom horizontal
(j) Check the hydraulic system including the pump at 10 percent above the pump manufacturer's maximum operating pressure setting for normal workload conditions.

* 4.4.11 Body treatment and painting. The certification regarding the cleaning, treating and prime painting, as required by MIL-STD-1223, shall be presented to Government representatives for examination and approval.

4.4.12 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample and be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.4.13 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under the contract until corrective action has been taken. In addition, subsequent vehicles that do not conform to the accepted production sample vehicle shall be rejected until corrected.

4.5 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.6 have been accomplished.

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5. PACKAGING

5.1 Vehicle processing. Vehicles shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general nontactical use by the Government for distribution lines not exceeding 100,000 volts with the basket liner installed for utility line maintenance, replacement of lamps, tree trimming, painting, window washing, and many other types of overhead construction and maintenance work.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, size, and model required (see 1.2).
(c) Identification of appropriate service for painting and marking (see 3.1.1.1).
(d) Concealed marking, if required (see 3.1.1.1).
(e) Rustproofing, if required (see 3.1.1.2).
(f) Tropical rustproofing, if required (see 3.1.1.2).
(g) Heavy duty cooling system, if required for model 4x2 (see
(h) Drive hubs, if required for model 4x4, classes A and E (see 3.1.1.8).
(i) Silicone brake fluid, if required (see 3.1.1.9).
(j) Gasoline engine, if required (see 3.4.1.2).
(k) Silicone hoses, if required (see 3.4.1.6).
(l) Power plant heaters and fuel warmer, if required (see 3.4.1.7).
(m) If a battery heater is not required (see 3.4.1.7).
(n) A measured shot type ether system, if required (see 3.4.2.1).
(o) Electric trailer brake controller, if required (see 3.4.2.7).
(p) Automatic or semiautomatic transmission, if required (see 3.4.5.3).
(q) Two-speed axle, if required for model 4x2 (see 3.4.9.1).
(r) Traction control, if required (see 3.4.9.2).
(s) Wide base tires, if required (see 3.4.10).
(t) Disc type wheels, if required (see 3.4.10).
(u) Bias ply tires, if required (see 3.4.10.1).
(v) If a spare tire carrier is not required (see 3.4.10.3).
(w) Spare wheel or rim, if required (see 3.4.10.4).
(x) Spare tire assembly, if required (see 3.4.10.5).

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(y) Full air brakes for classes A and E, if required (see 3.4.11.2).
(z) Automatic slack adjuster, if required (see 3.4.11.2(j)).
(aa) Trailer brake control system, if required (see 3.4.11.3).
(ab) Brake controls for use from a towing vehicle, if required (see 3.4.11.4).
(ac) Individual driver and passenger seats, if required (see 3.4.12).
(ad) Crew-cab, if required (see 3.4.12.1).
(ae) Tools, if required (see 3.4.16.1).
(af) Air horn, if required (see 3.4.19).
(ag) Front mounted winch, if required (see 3.4.21).
(ah) Reel for front mounted winch, if required (see 3.4.21.4).
(ai) Access steps, if required (see 3.5.1.2).
(aj) Water cask, if required (see 3.5.5.1).
(ak) Vise, if required (see 3.5.5.2).
(al) Portable gasoline engine driven generator, if required (see 3.5.5.3).
(am) Boom covers, if required (see 3.6.4.6).
(an) Non-conductive extension, if required (see 3.6.4.7).
(ao) Lifting attachment for conductors, if required (see 3.6.4.8).
(ap) If basket liner is not required (see 3.6.5.1.2).
(aq) Guard rail basket, if required (see 3.6.5.2).
(ar) Rotating baskets, if required (see 3.6.5.7).
(as) Single lever control at basket, if required (see 3.6.6).
(at) Spring lockouts, for type I, size 30 and 35 only, if required (see 3.6.7.2).
(au) Hydraulic fluid conforming to MIL-H-46170, if required (see 3.6.8.5).
(av) Rechargeable electrical system for driving hydraulic pump, if required (see 3.6.8.8).
(aw) Auxiliary engine for driving hydraulic pump, if required (see 3.6.9).
(ax) Auxiliary engine fuel tank, if required (see 3.6.9.6.1).
(ay) Air compressor, if required (see 3.6.10).
(az) If a basket hydraulic tool outlet and hose are not required (see 3.6.11.2).
(ba) Hydraulic capstan hoist, if required (see 3.6.11.4).
(bb) Hydraulic impact wrench and drill, if required (see 3.6.11.5).
(bc) Hydraulic chain saw, if required (see 3.6.11.6).
(bd) Hydraulic chain saw, pole mounted, if required (see 3.6.11.7).
(be) Hydraulic circular saw, pole mounted, if required (see 3.6.11.8).
(bf) Hydraulic pruner, pole mounted, if required (see 3.6.11.9).
(bg) Hydraulic hose reel, if required (see 3.6.11.12).

6.3 Performance prediction. Completed copies of SAE Truck Ability Prediction Procedure form and computation for low speed requirement for

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each vehicle model furnished under contract should be submitted as specified in the contract. The SAE Work Sheet Item 1 should include vehicle model number, wheelbase, and engine model number. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry, air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1	- Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	- Altitude Factor. 1.00
Table 3	- Rolling Factor. 1.613
Table 4	- Area Factor. 0.173
Table 5	- Velocity Factor. 250.0
Table 6	- Altitude Factor. 1.00
Table 7	- Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher, tabulated 1,000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 50 mph geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	- Grade Factor. 0.75
Table 8A	- Correction Factor. Not required.
Table 9	- Road Factor. 0.0.

6.4 Instructions. The requesting service should include provisions for the contractor to supply physical instructions to the using service on the proper operation of the equipment.

6.5 Basket insulation. The 30,000-volt basket insulation is required (see 3.6.3.1) to obtain a heavy duty basket devoid of thin areas and imperfections in the basket walls.

* 6.6 Identification of changes. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-45398H	TRUCKS MAINTENANCE: WITH 30 TO 60 FOOT MULTI-POSITION AERIAL PLATFOR, 15,000 TO 32,000 POUNDS GVW, 4x2 AND

3a. NAME OF SUBMITTING ORGANIZATION 4x4, COMMERCIAL.

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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